

Single-choice questions (○)

Q1. What is the purpose of an "activation function" in a neural network?

- ☐ To calculate the loss of the model
- ☐ To normalize the input data
- ☐ To introduce non-linearity, allowing the network to learn complex patterns
- ☐ To connect neurons from different layers

Q2. Which Huawei Cloud service is used for monitoring resource performance and setting up alarms?

- ☐ IAM
- ☐ Cloud Eye
- ☐ Cloud Trace
- ☐ FunctionGraph

Q3. In a Convolutional Neural Network (CNN), what is a "pooling" layer primarily used for?

- ☐ To add more filters to the network
- ☐ To increase the spatial dimensions of the feature maps
- ☐ To reduce the size of the feature maps, making the model more robust to variations
- ☐ To introduce more non-linearity

Q4. Which of the following is a key advantage of using Huawei's HiLens service for edge AI?

- ☐ It is cheaper to use than cloud-based AI for large-scale training.
- ☐ It processes data on the device, reducing latency and bandwidth usage.
- ☐ It requires a constant, high-speed internet connection to operate.
- ☐ It is only compatible with Python and PyTorch.

Q5. The process of packaging a model and all its dependencies into a single, portable unit for deployment is often referred to as:

- ☐ Batch processing
- ☐ Containerization
- ☐ Model training
- ☐ Data preparation

Q6. Which type of deep learning model is best suited for time series forecasting?

- ☐ Convolutional Neural Network (CNN)
- ☐ Recurrent Neural Network (RNN)
- ☐ Simple Perceptron
- ☐ Support Vector Machine (SVM)

Q7. Which MLOps principle focuses on ensuring that an AI model's performance does not degrade over time due to changes in real-world data?

- ☐ CI/CD
- ☐ Reproducibility
- ☐ Model drift detection
- ☐ Data validation

Q8. In ModelArts, which tool would you use to manage multiple versions of a trained model and its associated metrics?

- ☐ Notebooks
- ☐ Data Management
- ☐ Experiment Management
- ☐ ExeML

Q9. Which of the following best describes the role of a "feature" in a machine learning model?

- ☐ The final output or prediction of the model
- ☐ The algorithm used to train the model
- ☐ A measurable property or characteristic of the data
- ☐ The non-functional requirements of the system

Q10. What is the main difference between a functional and a non-functional requirement for an AI application?

- ☐ Functional requirements describe what the system does, while nonfunctional requirements describe how it performs.
- ☐ Functional requirements are for the developer, while non-functional requirements are for the user.
- ☐ Functional requirements are optional, while non-functional requirements are mandatory.
- ☐ There is no difference; they are interchangeable terms.

Multiple-choice questions (☐)

Q11. Which of the following statements about Recurrent Neural Networks (RNNs) are true?

- ☐ They are designed to handle sequential data.
- ☐ They have a "memory" that allows them to use information from previous steps.
- ☐ They are the best choice for image classification.
- ☐ They use convolutions to extract features.

Q12. What are some of the key capabilities of Huawei's ModelArts platform?

- ☐ Providing a managed environment for Jupyter notebooks
- ☐ Supporting popular deep learning frameworks like TensorFlow and PyTorch
- ☐ Offering a tool for automated machine learning (AutoML)
- ☐ Providing infrastructure for building a website

Q13. Which of the following are benefits of using an MLOps approach for AI development?

- ☐ Faster deployment of new models
- ☐ Automated retraining of models
- ☐ Improved collaboration between teams
- ☐ Eliminating the need for data preparation

Q14. In an MLOps pipeline, which services could be used for continuous integration and delivery?

- ☐ OBS for data and model storage
- ☐ ModelArts for automated training and deployment jobs
- ☐ FunctionGraph to trigger the pipeline
- ☐ VPC for network isolation

Q15. Which of the following are common use cases for Huawei's HiLens service?

- ☐ Retail store analytics (e.g., customer traffic)
- ☐ Smart city surveillance
- ☐ Logistics and inventory management
- ☐ Hosting a static website

Q16. What are some common challenges that can be addressed by MLOps?

- ☐ Model drift
- ☐ Manual and error-prone deployments
- ☐ Lack of collaboration between data science and engineering teams
- ☐ Insufficient computing power for training

Q17. Which of the following are key metrics used to evaluate a classification model's performance?

- ☐ Accuracy
- ☐ Latency
- ☐ Precision
- ☐ Recall

Q18. What are some of the steps involved in "error analysis" for a machine learning model?

- ☐ Identifying false positives and false negatives
- ☐ Examining the data points that the model misclassified
- ☐ Adjusting the model's hyper-parameters
- ☐ Gathering new data to improve the model

Q19. Which of the following services are part of the Huawei Cloud EI (Enterprise Intelligence) portfolio?

- ☐ OCR (Optical Character Recognition)
- ☐ NLP (Natural Language Processing)
- ☐ Computer Vision
- ☐ Cloud Backup and Recovery (CBR)

Q20. What are some of the advantages of using a hosted notebook environment like ModelArts Notebooks?

- ☐ Pre-installed deep learning frameworks
- ☐ Automatic resource scaling (CPU/GPU)
- ☐ Direct access to cloud storage services (like OBS)
- ☐ You do not need a Huawei Cloud account to use them.

Q21. Which of the following are considered types of model deployment?

- ☐ Real-time inference
- ☐ Batch inference
- ☐ On-device (edge) inference
- ☐ Serverless inference

Q22. Which of the following are true statements about the pandas library?

- ☐ It is built on top of NumPy.
- ☐ Its primary data structure is the DataFrame.
- ☐ It is optimized for numerical operations on large arrays.
- ☐ It can read and write various file formats like CSV and Excel.

Q23. In a deep neural network, what are the roles of the input and output layers?

- ☐ The input layer accepts the raw data.
- ☐ The output layer provides the final prediction.
- ☐ The input layer performs complex calculations.
- ☐ The output layer connects to the first hidden layer.

Q24. When performing a requirement analysis, what would you consider when determining if a project is suitable for an AI solution?

- ☐ The availability of sufficient, high-quality data.
- ☐ The presence of a clear problem that can be solved with a pattern-finding algorithm.
- ☐ The business value of the solution.
- ☐ The price of the AI service.

Q25. Which of the following are considered best practices for model testing and evaluation?

- ☐ Using a separate test set that the model has never seen before
- ☐ Analyzing misclassifications using a confusion matrix
- ☐ Re-training the model on the test set to improve accuracy
- ☐ Benchmarking the model's latency for production readiness

Q26. What does an "end-to-end" AI pipeline typically include?

- ☐ Data ingestion
- ☐ Model training
- ☐ Deployment
- ☐ Monitoring

Q27. Which of the following are aspects of the "Model Training Lifecycle" on ModelArts?

- ☐ Creating a dataset from OBS
- ☐ Configuring training parameters (e.g., learning rate)
- ☐ Submitting a training job to the cloud
- ☐ Registering a domain name for the AI application

Q28. A Convolutional Neural Network (CNN) is specifically designed to work with which type of data?

- ☐ Sequential data
- ☐ Tabular data
- ☐ Image and video data
- ☐ Text data

Q29. In a CI/CD pipeline, what would the "CI" (Continuous Integration) phase involve for an AI project?

- ☐ Merging code from multiple developers
- ☐ Running automated tests on the code and data
- ☐ Deploying a new model to production
- ☐ Training a model with new data

Q30. What is the purpose of an "API Gateway" in a serverless architecture?

- ☐ To store data in a highly scalable way
- ☐ To provide a single, managed entry point for API requests
- ☐ To execute serverless functions
- ☐ To monitor the performance of an application

Fill-in-the-blank questions ()

Q31. The process of automatically building a machine learning model without significant manual intervention is called _____.

Q32. In Huawei Cloud, the service that provides scalable, on-demand compute instances is _____.

Q33. To store and manage large files like datasets for ModelArts, you would use Huawei Cloud's _____ service.

Q34. The MLOps term for a model's performance degrading over time due to changes in data is _____.

Q35. The full name for the "HCCDA" certification is "Huawei Certified ICT _____ - Artificial Intelligence."

Q36. A serverless service that allows you to run code in response to events is called _____.

Q37. In a neural network, a single node or "neuron" is often referred to as a _____.

Q38. The testing method that compares the performance of two different models by directing live traffic to both is _____.

Q39. Huawei's service for managing and deploying AI on edge devices is _____.

Q40. In the ModelArts UI, the tool you would use to compare the performance metrics and loss curves of different training jobs is _____.

Match/Drag & Drop questions

Q41. Match the Huawei service to its MLOps role:

Service	Role	Answer
OBS	Monitoring and alarming	
ModelArts	Data and model storage	
Cloud Eye	Automated training and deployment	

Q42. Match the AI concept to its use case:

Concept	Case	Answer
Supervised Learning	Clustering customer data into segments	
Unsupervised Learning	Predicting house prices based on features	
Reinforcement Learning	A robot learning to navigate a maze	

Q43. Match the AI testing concept to its definition:

Concept	Definition	Answer
Precision	The ratio of correctly classified positive observations to all positive observations	
Recall	The ratio of correctly classified positive observations to all observations that are actually positive	
Accuracy	The ratio of correct predictions to total predictions	

Q44. Match the programming tool to its function:

Tool	Function	Answer
Pandas	Performing numerical operations	
NumPy	Making HTTP requests to an API	
requests	Loading and manipulating tabular data	

Q45. Match the type of neural network layer to its function:

Layer	Function	Answer
Convolutional Layer	Reduces the spatial size of the representation	
Pooling Layer	Applies a filter to an image to create a feature map	
Recurrent Layer	Processes sequential data by maintaining a hidden state	

Q46. Match the AI API to its application:

API	Application	Answer
NLP API	Transcribing spoken words into text	
OCR API	Extracting text from images and documents	
Speech API	Analyzing the emotional tone of a text review	

Q47. Match the MLOps challenge to its solution:

Challenge	Solution	Answer
Model Drift	Using a confusion matrix to analyze misclassifications	
Error Analysis	Automating the retraining pipeline	
Slow Deployment	Using a CI/CD pipeline	

Q48. Match the Huawei Cloud service to its category:

Service	Category	Answer
OBS	Networking	
VPC	Storage	
FunctionGraph	Serverless	

Q49. Match the ModelArts feature to its use case:

Feature	Use Case	Answer
AutoML	Comparing two different training runs	
Notebook	Building an end-to-end pipeline	
Experiment Management	Training a model without manual hyperparameter tuning	

Q50. Match the AI/ML term to its definition:

Term	Definition	Answer
Feature Engineering	The process of selecting and transforming raw data into features	
Hyper-parameter Tuning	Adjusting parameters that control the training process	
Overfitting	When a model performs well on training data but poorly on new data	

True/False questions

Q51. The primary function of a CNN is to process sequential data like text.

- ☐ True
- ☐ False

Q52. A non-functional requirement is just as important as a functional requirement for a successful AI project.

- ☐ True
- ☐ False

Q53. ModelArts supports both TensorFlow and PyTorch frameworks for model training.

- ☐ True
- ☐ False

Q54. Model drift can be detected by monitoring the performance metrics of a model in production.

- ☐ True
- ☐ False

Q55. The Huawei HiLens service requires all data to be sent to the cloud for inference.

- ☐ True
- ☐ False

Q56. In the context of MLOps, CI/CD stands for "Continuous Integration/Continuous Deployment."

- ☐ True
- ☐ False

Q57. The requests library in Python is primarily used for machine learning.

- ☐ True
- ☐ False

Q58. A confusion matrix can help you identify false positives and false negatives in a classification task.

- ☐ True
- ☐ False

Q59. A single perceptron can learn to solve complex, non-linear classification problems.

- ☐ True
- ☐ False

Q60. Huawei's Cloud Eye service can be configured to trigger alarms when an AI model's CPU utilization exceeds a certain threshold.

- ☐ True
- ☐ False

Single-choice questions (○)

Q1. Which Huawei service is primarily responsible for managing user permissions, roles, and access policies?

- ☐ Cloud Eye
- ☐ IAM
- ☐ VPC
- ☐ SWR

Q2. The "CI" in CI/CD stands for:

- ☐ Cloud Infrastructure
- ☐ Continuous Integration
- ☐ Cloud Intelligence
- ☐ Continuous Improvement

Q3. What is the main purpose of a "security group" in a Huawei Cloud Virtual Private Cloud (VPC)?

- ☐ To manage user login credentials
- ☐ To act as a virtual firewall for ECS instances
- ☐ To monitor network latency
- ☐ To provide DNS resolution for private IPs

Q4. When deploying a model as a real-time API endpoint, what is the main benefit?

- ☐ Low-latency inference for single requests
- ☐ High-throughput processing for large datasets
- ☐ Reduced cost compared to batch deployment
- ☐ It eliminates the need for a deep learning framework

Q5. Which of the following is a key component of a successful MLOps strategy?

- ☐ Manual, ad-hoc model deployment
- ☐ Automating the model retraining process
- ☐ Using only one type of deep learning framework
- ☐ Avoiding version control for code and data

Q6. The Huawei Cloud service used for storing container images and managing their versions is:

- ☐ OBS
- ☐ SFS
- ☐ CCE
- ☐ SWR

Q7. Which of the following is a core characteristic of a serverless architecture like Huawei's FunctionGraph?

- ☐ You must manually provision and manage servers.
- ☐ You are billed for idle compute time.
- ☐ You only pay for the compute time you consume.
- ☐ It is only suitable for large, long-running processes.

Q8. When analyzing model performance, what does "latency" refer to?

- ☐ The time it takes to train the model
- ☐ The delay between receiving a request and returning a prediction
- ☐ The total number of predictions made in an hour
- ☐ The accuracy of the model's predictions

Q9. In the context of a deep learning model, what is "overfitting"?

- ☐ When a model is too simple to learn from the data
- ☐ When a model learns a general trend from the data
- ☐ When a model performs well on training data but poorly on unseen data
- ☐ When a model trains too slowly

Q10. What is the primary role of an "API Gateway" in an AI application architecture?

- ☐ To store model training data
- ☐ To serve as a single entry point for all API requests
- ☐ To run a deep learning training job
- ☐ To manage the security of individual servers

Multiple-choice questions (☐)

Q11. Which of the following are benefits of using a private network (VPC) for your AI applications?

- ☐ Network isolation and security
- ☐ Control over IP address ranges and subnets
- ☐ Ability to host a public website
- ☐ Direct access to all other users' cloud resources

Q12. What are some of the key features of Huawei Cloud's IAM service?

- ☐ Creating users and user groups
- ☐ Assigning fine-grained permissions to users
- ☐ Enabling multi-factor authentication (MFA)
- ☐ Deploying a machine learning model

Q13. Which of the following are examples of security best practices in Huawei Cloud?

- ☐ Using IAM roles with the principle of least privilege
- ☐ Storing all credentials directly in code
- ☐ Enabling Cloud Eye for monitoring
- ☐ Using security groups to control network access

Q14. What are some common use cases for a serverless function like FunctionGraph?

- ☐ Resizing an image after it's uploaded to OBS
- ☐ Triggering a model retraining pipeline
- ☐ Handling a spike in web traffic to an API endpoint
- ☐ Hosting a long-running database service

Q15. Which of the following are key steps in a basic MLOps CI/CD pipeline?

- ☐ A new training dataset is uploaded to OBS.
- ☐ The pipeline is manually triggered by a user.
- ☐ A new model is trained and evaluated.
- ☐ The updated model is automatically deployed to a production endpoint.

Q16. What are some of the components involved in a model's "testing and evaluation" phase?

- ☐ A separate test dataset
- ☐ A confusion matrix to analyze results
- ☐ Performance metrics like accuracy and latency
- ☐ The final production deployment

Q17. Which of the following are key considerations when designing an AI application architecture?

- ☐ Latency requirements
- ☐ Data security and privacy
- ☐ Scalability and cost
- ☐ The color scheme of the user interface

Q18. The Huawei Cloud service that helps with "continuous integration" and "continuous delivery" for code is:

- ☐ DevCloud
- ☐ OBS
- ☐ ModelArts
- ☐ FunctionGraph

Q19. Which of the following are examples of how pandas is used in a typical AI project workflow?

- ☐ Loading raw data from a CSV file
- ☐ Cleaning and pre-processing data
- ☐ Visualizing data using built-in plotting functions
- ☐ Training a deep learning model directly

Q20. What are some benefits of using a requests library in Python for API calls?

- ☐ It simplifies the process of sending HTTP requests.
- ☐ It can automatically handle authentication.
- ☐ It makes it easier to handle different response formats.
- ☐ It is faster than using NumPy for numerical operations.

Q21. Which of the following are responsibilities of an MLOps engineer?

- ☐ Designing the model architecture
- ☐ Building and maintaining automated pipelines
- ☐ Monitoring model performance in production
- ☐ Identifying and addressing model drift

Q22. A robust AI application architecture should be:

- ☐ Scalable
- ☐ Secure
- ☐ Highly available
- ☐ Dependent on manual intervention

Q23. In the context of MLOps, which of the following are methods to ensure "reproducibility"?

- ☐ Versioning datasets
- ☐ Versioning model code and configurations
- ☐ Documenting the training environment and dependencies
- ☐ Manually creating new training jobs each time

Q24. Which of the following are types of access keys used for API authentication on Huawei Cloud?

- ☐ Access Key ID
- ☐ Secret Access Key
- ☐ API Key
- ☐ Personal Access Token

Q25. Which of the following are key features of the Huawei HiLens SDK?

- ☐ Building and packaging a simple AI model
- ☐ Deploying the model to a HiLens device
- ☐ Running a full-scale deep learning training job
- ☐ Monitoring the status of the edge device

Q26. What does an "end-to-end" AI application lifecycle refer to?

- ☐ The full process from data collection to model monitoring
- ☐ The process of training a model only
- ☐ The process of deploying a model to a production environment
- ☐ The cycle of a user interacting with the application

Q27. Which of the following are important for securing your Huawei Cloud account?

- ☐ Using a strong password
- ☐ Enabling Multi-Factor Authentication (MFA)
- ☐ Not sharing your Access Key/Secret Key
- ☐ Publishing your access keys in a public repository

Q28. What is the purpose of a "test set" in machine learning?

- ☐ To train the model on a small portion of the data
- ☐ To evaluate the model's performance on data it has never seen
- ☐ To tune the model's hyper-parameters
- ☐ To clean and pre-process the data before training

Q29. When using a Huawei Cloud API, what are some potential errors that you might encounter?

- ☐ Authentication failure (wrong AK/SK)
- ☐ Rate limit exceeded
- ☐ Incorrect input format
- ☐ Disk space issues on your local machine

Q30. A well-architected AI solution should be:

- ☐ Cost-optimized
- ☐ Reliable
- ☐ Operationally excellent
- ☐ Monolithic and difficult to update

Fill-in-the-blank questions (___)

Q31. The Huawei service that helps you manage identity and access to your cloud resources is _____.

Q32. The MLOps practice of tracking and documenting all changes to a model's code, data, and configurations is called _____.

Q33. A network component that acts as a virtual firewall for your ECS instances is a _____.

Q34. When a model's accuracy on new data is significantly worse than its accuracy on training data, the model is _____.

Q35. In a serverless architecture, you only pay for the compute time you _____.

Q36. The Huawei Cloud service for storing container images is _____.

Q37. The practice of managing the AI model lifecycle from development to production is known as _____.

Q38. A measurable property or characteristic of data used for training a machine learning model is called a _____.

Q39. In Huawei Cloud's API authentication, the Secret Access Key (SK) is a secret string that you must keep _____.

Q40. The process of packaging a model and its dependencies into a single, portable unit is _____.

Match/Drag & Drop questions

Q41. Match the MLOps concept to its definition:

Concept	Definition	Answer
Reproducibility	The process of automatically testing and deploying models	
CI/CD	Ensuring a model can be rebuilt with the same results	
Model Drift	A model's performance degrading over time	

Q42. Match the security best practice to its goal:

Best Practice	Goal	Answer
Principle of Least Privilege	Only giving users the permissions, they absolutely need	
Security Groups	Adding an extra layer of security to user logins	
Using MFA	Controlling inbound and outbound network traffic	

Q43. Match the AI testing tool to its purpose:

Tool	Purpose	Answer
Performance Benchmarking	Measuring the speed of a model's predictions	
A/B Testing	Analyzing the types of errors a classification model makes	
Confusion Matrix	Comparing two different models in a live environment	

Q44. Match the Huawei service to its function:

Service	Function	Answer
OBS	Container image storage	
IAM	Identity and Access Management	
SWR	Object-based storage	

Q45. Match the AI term to its definition:

Term	Definition	Answer
Dataset	A measurable property of the data	
Label	The output or target variable you are trying to predict	

Feature	A collection of data used for training and testing	
---------	--	--

Q46. Match the deployment type to its use case:

Deployment Type	Use Case	Answer
On-device	Processing millions of images overnight	
Batch	A chatbot responding to a user's question	
Real-time	Object detection on a drone with limited connectivity	

Q47. Match the Huawei Cloud service to its category:

Service	Category	Answer
ModelArts	AI Service	
ECS	Compute Service	
Cloud Eye	Monitoring Service	

Q48. Match the programming concept to its role in AI:

Concept	Role	Answer
Loops	Used to repeat a block of code, e.g., for processing each row of data	
Variables	Used to define reusable blocks of code for specific tasks	
Functions	Used to store and manipulate data, e.g., a dataset or a single number	

Q49. Match the AI lifecycle stage to its activity:

Stage	Activity	Answer
Data Preparation	Collecting, cleaning, and labeling data	
Training	Running an algorithm to learn from the data	
Deployment	Making the model available for use	

Q50. Match the neural network component to its function:

Component	Function	Answer
Input Layer	The final layer that produces the prediction	
Output Layer	The layer between the input and output layers	
Hidden Layer	The first layer that receives the raw data	

True/False questions

Q51. The principle of least privilege dictates that users should have maximum access to all resources.

- ☐ True
- ☐ False

Q52. A confusion matrix is a good tool for evaluating the performance of a regression model.

- ☐ True
- ☐ False

Q53. ModelArts provides a managed CI/CD pipeline for automating the AI model lifecycle.

- ☐ True
- ☐ False

Q54. Model drift occurs when the data distribution in production changes, causing the model to perform worse.

- ☐ True
- ☐ False

Q55. All Huawei Cloud APIs can be accessed using a RESTful API call.

- ☐ True
- ☐ False

Q56. In an MLOps pipeline, model retraining should be triggered manually.

- ☐ True
- ☐ False

Q57. A serverless function like FunctionGraph is ideal for long-running, CPUintensive tasks.

- ☐ True
- ☐ False

Q58. A well-architected AI solution should prioritize performance over security.

- ☐ True
- ☐ False

Q59. The Huawei HiLens service is an example of a serverless AI solution.

- ☐ True
- ☐ False

Q60. The IAM service helps you manage your budget and billing on Huawei Cloud.

- ☐ True
- ☐ False